



MCKV INSTITUTE OF ENGINEERING

NAAC Accredited "A" Grade Autonomous Institute under UGC Act 1956
Approved by AICTE & affiliated to Maulana Abul Kalam Azad University of Technology, West Bengal
243 G.T. Road (N), Liluah, Howrah- 711204, West Bengal, India
Ph: +91 33 26549315/17 Fax +91 33 26549318 Web: www.mckvie.edu.in/

Curriculum for Undergraduate Degree (B. Voc.) in Software Development (w.e.f. AY: 2022-23)

Part III: Detailed Curriculum

Fifth Semester (Third Year)

Course Name:	Multimedia Technology		
Course Code:	BSD501		
Semester:	V	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Concepts of Graphics and Visualization
Full Marks:	50		
Examination Scheme:	Semester Examination: 35	Continuous Assessment:10	Attendance: 05

Course Objectives:	
1.	To introduce various aspects of multimedia components like image, text, audio, video, animation and computer graphics.
2.	To allow students gain knowledge so as to exercise hands-on work later on through a series of practical skill-building tasks.
3.	To encourage an individual's creativity which will also arouse the interest in the course and further motivate towards developing one's career in the area of multimedia and related applications.

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Introduction To Multimedia, Needs and Areas of use, Identifying Multimedia Elements - Text, Images, Graphics, Sound, Animation and Video, Making a simple Multimedia application With PowerPoint. TEXT - Concepts of Plain & Formatted Text, RTF & HTML Texts, Using Common Text Preparation Tools, Conversion to and from of Various Text Formats, Creating text using standard software.	8L
2.	SOUND - Sound and its Attributes, Sound and Its Effects in Multimedia, Frequency, Sound Depth, Channels and its Effects on Quality and Storage, Size Estimation of Space of a Sound File, MIDI and MP3 Files and Devices, Recording and editing sound using any sound editor like Audacity, Sound forge etc.	4L
3.	IMAGES - Importance of Images Graphics in Multimedia, Vector and Raster Graphics, Regular Graphics vs. Interlaced Graphics, Image Capturing Methods - Scanner, Digital Camera etc. Color models-RGB, CYMK, Hue, Saturation, and Brightness, Various Attributes of Images Size, Color, Depth etc. Various Image File Format BMP, JPG, GIF, PNG,	10L



MCKV INSTITUTE OF ENGINEERING

NAAC Accredited "A" Grade Autonomous Institute under UGC Act 1956
Approved by AICTE & affiliated to Maulana Abul Kalam Azad University of Technology, West Bengal

243 G.T. Road (N), Liluah, Howrah- 711204, West Bengal, India

Ph: +91 33 26549315/17 Fax +91 33 26549318 Web: www.mckvie.edu.in/

	and TIFF Format Their Features And Limitations, Image format conversion, various effects on images. Create images using Photoshop, CorelDraw and apply various effects, Using Layers, Channels and Masks in images.	
4	VIDEO- Basics of Video, Analog and Digital Video Type of Video, Digitization of Analog Video, Video Standard – NTSC, Pal, HDTV, Digital file Formats AVI, MKV, MPEG, Video Editing and Movie Making Tools, Converting formats of videos, recording and editing videos using video editing software like adobe premiere or other.	8L
5	ANIMATION- Concepts of animation, 2D and 3D animation, tools for creating animation, character and text animation, creating simple animation using Adobe Flash, Creating multimedia package using all media components.	6L
Total		36L

Course Outcomes:

After completion of the course, students will be able to:

1	Express basic concepts related to multimedia like : a. explain basic use approaches regarding multimedia applications. b. categorize multimedia applications according to the purpose of use. c. explain the benefits of multimedia for instructional settings. d. list limitations of multimedia applications regarding design and development.
2.	Explain types of multimedia applications like : a. explain audio-based multimedia products. b. explain visual-based multimedia products. c. explain animation-based multimedia products.
3.	Design multimedia application to conduct planning towards some context specific multimedia product.
4.	Integrate multimedia applications to instructional settings like : a. relate & compare a developed multimedia application with a typical instructional software. b. narrate sample instructional activities into a pure multimedia application.

Learning Resources:

1.	Multimedia & Its Applications, V.K. Jain, Khanna Publishing House
2.	Fundamentals of Multimedia, Ramesh Bangia, Khanna Publishing House
3.	Principles of Multimedia, Ranjan Parekh, Mc Graw Hill Education



MCKV INSTITUTE OF ENGINEERING

NAAC Accredited "A" Grade Autonomous Institute under UGC Act 1956
Approved by AICTE & affiliated to MaulanaAbulKalam Azad University of Technology, West Bengal
243 G.T. Road (N), Liluah, Howrah- 711204, West Bengal, India
Ph: +91 33 26549315/17 Fax +91 33 26549318 Web: www.mckvie.edu.in/

Course Name:	Computer Network and System Administration		
Course Code:	BSD502		
Semester:	V	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Knowledge of Operating System
Full Marks:	50		
Examination Scheme:	Semester Examination: 35	Continuous Assessment:10	Attendance: 05

Course Objectives:	
1.	Learning about computer network organization and implementation.
2.	Obtaining a theoretical understanding of data communication and computer networks, and gaining practical experience in installation, monitoring, and troubleshooting of current LAN systems
3.	Students are introduced to computer communication network design and its operations.

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1.	Introduction to computer Network - Communication: An Essential Part of Our Lives, Communicating in a Network-Centric World, Network as a Platform, Architecture of the Internet, Trends in Networking. Network Concept, Benefits of Network, Network classification (PAN, LAN, MAN, WAN), Peer to Peer, Client Server architecture, Transmission media: Guided & Unguided, Network Topologies.	5L
2.	Application Layer Functionality and Protocols - Applications: The Interface Between the Networks, Making Provisions for Applications and Services, Application Layer Protocols and Services Examples. OSI Transport Layer - Roles of the Transport Layer, IPv4 Addresses, TCP: Communicating with Reliability, UDP: Communicating with Low Overhead.	7L
3.	OSI Network Layer - IPv4, Networks: Dividing Hosts into Groups, Routing, How Data Packets Are Handled, Routing Processes. Addressing the Network- IPv4 Addresses for Different Purposes, Assigning Addresses, Calculating Addresses, Testing the Network Layer OSI Data Link Layer - Data Link Layer, MAC Techniques, MAC Addressing and Framing Data	10L
4.	Define Name resolution, Concept of DNS Installation and Configuration. DNS Client, Manage and Troubleshoot DNS. Concept of Dynamic Host Configuration Protocol. Concept of FTP and its application.	10L
5.	Backup and Restore: Requirement for Backup and Recovery. Issue for Backup and Recovery	4L
Total		36L



MCKV INSTITUTE OF ENGINEERING

NAAC Accredited "A" Grade Autonomous Institute under UGC Act 1956
Approved by AICTE & affiliated to Maulana Abul Kalam Azad University of Technology, West Bengal

243 G.T. Road (N), Liluah, Howrah- 711204, West Bengal, India

Ph: +91 33 26549315/17 Fax +91 33 26549318 Web: www.mckvie.edu.in/

Course Outcomes:

After completion of the course, students will be able to:

- | | |
|----|--|
| 1. | Analyse requirements and design network architecture for a given scenario. |
| 2. | Design and configure a client-server network and required network services for a given scenario. |
| 3. | Design and configure IP addressing schemes for a given scenario. |
| 4. | Manage system administration and data recovery. |

Learning Resources:

- | | |
|----|--|
| 1. | An Integrated Approach to Computer Networks, Bhavneet Sidhu, Khanna Publishing House |
| 2. | Computer Networks, Gupta & Gupta, Khanna Publishing House |
| 3. | Data Communications and Networking, Behrouz A. Forouzan, Mc-Graw Hill |
| 4. | Systems and Network Administration, A handbook for beginners, FIROSA. |



MCKV INSTITUTE OF ENGINEERING

NAAC Accredited "A" Grade Autonomous Institute under UGC Act 1956
Approved by AICTE & affiliated to Maulana Abul Kalam Azad University of Technology, West Bengal

243 G.T. Road (N), Liluah, Howrah- 711204, West Bengal, India

Ph: +91 33 26549315/17 Fax +91 33 26549318 Web: www.mckvie.edu.in/

Course Name:	Introduction to Biometrics		
Course Code:	BSD503		
Semester:	V	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	
Full Marks:	50		
Examination Scheme:	Semester Examination: 35	Continuous Assessment:10	Attendance: 05

Course Objectives:	
1.	Develop scientific foundation for design and implementation biometric identification.
2.	Provide Knowledge to evaluate biometric identification system.

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1.	Concepts - biometric recognition, biometrics, requirements for biometrics Biometric systems, their modes and architectures. Biometric system errors and evaluation.	14L
2.	Overview, comparison and evaluation of various biometrics. Unimodal biometric systems, their advantages, disadvantages and limits. Multimodal biometric systems, their modes of operation, levels of fusion.	10L
3.	Biometric pattern recognition methods. Privacy protection and social acceptance. Biometric standardization, data formats. Design and implementation of biometric systems, applications of biometric systems, biometric databases, security of biometric systems.	12L
Total		36L

Course Outcomes:	
After completion of the course, students will be able to:	
1.	Understand the technological modification of current biometric system.
2.	Gain knowledge in fields like Pattern Recognition, Image Processing etc.
3.	Evaluate and Design security systems with biometrics.

Learning Resources:	
1.	Biometrics: Concepts And Applications, G. R. Sinha, Sandeep B. Patil, Wiley
2.	Biometric Systems-Technology Design and Performance Evaluation, James Wayman, Anil Jain, David Maltoni, Dario Maio (Eds), Springer International Edition



MCKV INSTITUTE OF ENGINEERING

NAAC Accredited "A" Grade Autonomous Institute under UGC Act 1956
 Approved by AICTE & affiliated to Maulana Abul Kalam Azad University of Technology, West Bengal
 243 G.T. Road (N), Liluah, Howrah- 711204, West Bengal, India
 Ph: +91 33 26549315/17 Fax +91 33 26549318 Web: www.mckvie.edu.in/

Course Name:	Network Security and Cryptography		
Course Code:	BSD504		
Semester:	V	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Concepts of Computer Network
Full Marks:	50		
Examination Scheme:	Semester Examination: 35	Continuous Assessment:10	Attendance: 05

Course Objectives:	
1.	To understand the need for network and web security.
2.	To analyze the different symmetric and asymmetric key algorithms.
3.	To understand Internet security protocols and implement it in web design.
4.	To study the importance of firewall and different security challenges of mobile and web applications.

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Introduction, Need for Security, Security approaches, Principles of Security, Types of attacks.	5L
2.	Introduction, Plaintext & Cipher text, Substitution Techniques Transposition Techniques, Encryption & Decryption, Symmetric & Asymmetric key, Cryptography, Key Range & Key Size.	7L
3.	) Algorithm types & Modes, Overview of Symmetric Key Cryptography, DES (Data Encryption Standard) algorithm, IDEA (International Data Encryption Algorithm) algorithm, RC5 (Rivest Cipher 5). Algorithm at the preliminary level only. (Detailed of Algorithm not required).	8L
4.	Asymmetric Key Algorithm, Digital Signature and RSA - Introduction, Overview of Asymmetric key Cryptography, RSA algorithm, Symmetric & Asymmetric key Cryptography together, Digital Signature, Basic concepts of Message Digest and Hash Function (Algorithms on Message Digest and Hash function not required).	5L
5.	Internet Security Protocols, User Authentication - Basic Concepts, SSL protocol, HTTPS and HTTP, Designing A secured web site.	5L
6.	E-mail security, Firewall, Security challenges posted by mobile devices, Attack on mobile/cell phones, Theft, virus, Hacking, Bluetooth.	6L
Total		36L



MCKV INSTITUTE OF ENGINEERING

NAAC Accredited "A" Grade Autonomous Institute under UGC Act 1956
Approved by AICTE & affiliated to MaulanaAbulKalam Azad University of Technology, West Bengal

243 G.T. Road (N), Liluah, Howrah- 711204, West Bengal, India

Ph: +91 33 26549315/17 Fax +91 33 26549318 Web: www.mckvie.edu.in/

Course Outcomes:

After completion of the course, students will be able to:

- | | |
|----|--|
| 1. | To identify different types of security and attacks on computer network. |
| 2. | To differentiate between symmetric and asymmetric cryptography. |
| 3. | To identify different Internet security protocols and use that in secured website development. |
| 4. | To Identify different security threats related to e-mail application and mobile devices. |

Learning Resources:

- | | |
|----|---|
| 1. | Information & Computer Security, Sarika Gupta, Khanna Publishing House |
| 2. | An Integrated Approach to Computer Networks, Bhavneet Sidhu, Khanna Publishing House |
| 3. | Cryptography & Network Security: Atul Kahate, TMH. |
| 4. | "Cryptography and Network Security," William Stallings, 2nd Edition, Pearson Education Asia |



MCKV INSTITUTE OF ENGINEERING

NAAC Accredited "A" Grade Autonomous Institute under UGC Act 1956
Approved by AICTE & affiliated to Maulana Abul Kalam Azad University of Technology, West Bengal

243 G.T. Road (N), Liluah, Howrah- 711204, West Bengal, India

Ph: +91 33 26549315/17 Fax +91 33 26549318 Web: www.mckvie.edu.in/

Course Name:	Multimedia Lab		
Course Code:	BSD591		
Semester:	IV	Credit:	1.5
L-T-P:	0-0-3	Pre-Requisites:	
Full Marks:	50		
Examination Scheme:	Semester Examination: 30	Continuous Assessment:20	

Course Objectives:	
1.	To make the students understand and apply the graphics concepts and to develop, design and implement them using the respective tools.
2.	To understand the multimedia standards and apply them.
3.	To understand and apply the different multimedia tools seamlessly and interchangeably.

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1.	Image editing with Photoshop : A. Image editing : a. Create an image with multiple layers from different sources b. Alter the image properties like : pixel resolution, dimension (aspect ratio) c. Text insertion in the image d. Applying image background etc. B. Image restoration of an old and partially damaged b/w image and save it as a revived colour image. C. Poster / Banner design e.g. <i>Create an advertisement poster of 6' (width)×4'(length) promoting a product/event/enterprise with some suitable coloured texture(backdrop), images, graphics, promotional texts etc. Render the image into JPG/PNG format with a final resolution of 150 ppi/dpi</i> D. Logo design e.g. <i>Create a circular logo of size 1.5"×1.5" for any organization with transparent background. The logo may/ may not hold a text piece. Should be rendered in interlaced PNG format and should, in a way, reflect the organization's operations.</i>	21P
2.	Animation with Flash : A. Tween-ed animation for various drawn object using motion tween and shape tween with multiple layers B. Frame by frame animation using symbols etc. C. Masking animation D. Animation under a guided path E. Animation with text pieces	15P
Total		36P



MCKV INSTITUTE OF ENGINEERING

NAAC Accredited "A" Grade Autonomous Institute under UGC Act 1956
Approved by AICTE & affiliated to MaulanaAbulKalam Azad University of Technology, West Bengal

243 G.T. Road (N), Liluah, Howrah- 711204, West Bengal, India

Ph: +91 33 26549315/17 Fax +91 33 26549318 Web: www.mckvie.edu.in/

Course Outcomes:

After completion of the course, students will be able to:

- | | |
|----|--|
| 1. | Design an image using image-editing tools and apply effectively to the context given. |
| 2. | Create animated sequence with titles applying the principles of animation. |
| 3. | Apply acquired knowledge in the field of multimedia for product of a purpose like an advertisement or webpages in practice and independently continue to expand knowledge in this field. |

Learning Resources:

- | | |
|----|---|
| 1. | Mastering Photoshop, Web Tech Solutions Inc. |
| 2. | Learning Macromedia Flash, Suzanne Weixel, BPB Publications |
| 3. | Learning Dreamweaver CS4, Ramesh Bangia, Khanna Publishers |



MCKV INSTITUTE OF ENGINEERING

NAAC Accredited "A" Grade Autonomous Institute under UGC Act 1956
Approved by AICTE & affiliated to Maulana Abul Kalam Azad University of Technology, West Bengal

243 G.T. Road (N), Liluah, Howrah- 711204, West Bengal, India

Ph: +91 33 26549315/17 Fax +91 33 26549318 Web: www.mckvie.edu.in/

Course Name:	Computer Network and System Administration Lab		
Course Code:	BSD592		
Semester:	IV	Credit:	1.5
L-T-P:	0-0-3	Pre-Requisites:	Concept of Computer Network
Full Marks:	50		
Examination Scheme:	Semester Examination: 30	Continuous Assessment:20	

Course Objectives:	
1.	To identify different network devices.
2.	To provide idea of wired LAN.
3.	To provide knowledge of FTP, DNS and their utilities.

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1.	Familiarization with network devices: Ethernet switch, UTP cable, RJ-45 connector, UTP cable, I/O box, LAN tester, Patch panel etc.	6P
2.	Colour code of CAT5/CAT6 cable, preparation of straight through and cross over cable. Testing of cable by LAN tester. Connecting of systems with switches.	9P
3.	Setup and configuration of LAN (Windows). Demonstration of sharing files, folders (Different access modes) and printers. Idea of basic networking commands like: ping, ipconfig, ifconfig, traceroute etc.	6P
4.	User creation (Windows and Linux). Demonstration FTP configuration and its services.	9P
5.	Demonstration of DNS configuration and its services.	6P
Total		36P

Course Outcomes:	
After completion of the course, students will be able to:	
1.	Prepare patch cord to connect desktop, laptop, switch etc. in a network.
2.	Setup a LAN and sharing resources in an organization.
3.	Configure FTP, DNS and to implement services.

Learning Resources:	
1.	Internet And Networking Maintenance, G. P. Sangia, Atlantic Publishers & Distributors Pvt Ltd.
2.	Computer Systems And Networking Guide, Hans Weber



MCKV INSTITUTE OF ENGINEERING

NAAC Accredited "A" Grade Autonomous Institute under UGC Act 1956
Approved by AICTE & affiliated to MaulanaAbulKalam Azad University of Technology, West Bengal

243 G.T. Road (N), Liluah, Howrah- 711204, West Bengal, India

Ph: +91 33 26549315/17 Fax +91 33 26549318 Web: www.mckvie.edu.in/

Course Name:	On Job Training		
Course Code:	OJT581		
Semester:	IV	Credit:	15
L-T-P:	Sessional	Pre-Requisites:	
Full Marks:	200		
Examination Scheme:	Training in Semester: 200		

Training Scheme:

Students will go for an industrial training in the semester end for one month. After completion of the training, they will prepare a report and provide a presentation on the training in front of faculty members. On the basis of their report and presentation they will be evaluated.

Serial No.	Role of trainee
1	System Administrator and Network Engineer
2	Photo Editor and Animation Developer

NB: Any one of the above roles.